Car Rental Fleet Manager

Complete Java source code

```
import java.util.*;
import java.io.*;

class Car {
    private int id;
    private String name;
    private double rent;
    private boolean available;

    Car(int carId, String carName, double carRent) {
        id = carId;
        name = carName;
        rent = carRent;
        available = true;
    }

    int getId() {
        return id;
    }

    String getName() {
        return name;
    }

    boolean isAvailable() {
        return available;
    }

    void display() {
        System.out.println(id + " | " + name + " | Rs." + rent
                + " | " + (available ? "Available" : "Rented"));
    }

    void rentCar() throws Exception {
        if (available) {
            available = false;
            System.out.println(name + " rented successfully.");
        } else {
            throw new Exception(name + " is already rented.");
        }
    }

    void returnCar() throws Exception {
        if (!available) {
            available = true;
            System.out.println(name + " returned successfully.");
        } else {
            throw new Exception(name + " was not rented.");
        }
    }

    public String toString() {
        return id + " | " + name + " | Rs." + rent
                + " | " + (available ? "Available" : "Rented");
    }
}

public class CarRental {

    static ArrayList<Car> cars = new ArrayList<>();

    static void displayCars() {
        System.out.println("\\n--- CAR LIST ---");
        for (Car car : cars) {
            car.display();
        }
    }

    static Car findCar(int id) {
        for (Car car : cars) {
            if (car.getId() == id) {
                return car;
            }
        }
        return null;
    }

    // Recursive method to count available cars
    static int countAvailable(int index) {
        if (index == cars.size()) {
```

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        return 0;
    }

    int count = cars.get(index).isAvailable() ? 1 : 0;
    return count + countAvailable(index + 1);
}

static void searchCar(int id) {
    Car car = findCar(id);

    if (car != null) {
        car.display();
    } else {
        System.out.println("Car not found.");
    }
}

static void sortCars() {
    cars.sort(Comparator.comparing(Car::getName));
    System.out.println("Cars sorted by name.");
    displayCars();
}

static void showAvailableCars() {
    System.out.println("\\n--- AVAILABLE CARS ---");

    cars.stream()
        .filter(Car::isAvailable)
        .forEach(System.out::println);
}

static void saveCars() {
    try (PrintWriter out = new PrintWriter("cars.txt")) {
        for (Car car : cars) {
            out.println(car);
        }
        System.out.println("Fleet saved to cars.txt.");
    } catch (IOException e) {
        System.out.println("Error saving file.");
    }
}

static void readCars() {
    try (Scanner file = new Scanner(new File("cars.txt"))) {
        System.out.println("\\n--- SAVED FLEET ---");

        while (file.hasNextLine()) {
            System.out.println(file.nextLine());
        }
    } catch (FileNotFoundException e) {
        System.out.println("File not found. Save the fleet first.");
    }
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    // Module 3: Array used to initialize the fleet
    Car[] initialCars = {
        new Car(101, "GTR", 5000),
        new Car(102, "Porsche", 4500),
        new Car(103, "Bugatti", 3000)
    };

    cars.addAll(Arrays.asList(initialCars));

    int choice = 0;

    do {
        System.out.println("\\n===== CAR RENTAL FLEET MANAGER =====");
        System.out.println("1. View Cars");
        System.out.println("2. Rent Car");
        System.out.println("3. Return Car");
        System.out.println("4. Count Available Cars");
        System.out.println("5. Search Car");
        System.out.println("6. Sort Cars");
        System.out.println("7. Show Available Cars");
        System.out.println("8. Save Fleet");
        System.out.println("9. Read Saved Fleet");
        System.out.println("10. Exit");
        System.out.print("Enter your choice: ");

        try {
            choice = sc.nextInt();

            switch (choice) {
```

```
                    case 1:
                        displayCars();
                        break;

                    case 2:
                    case 3:
                        System.out.print("Enter car ID: ");
                        int id = sc.nextInt();
                        Car car = findCar(id);

                        if (car == null) {
                            System.out.println("Car not found.");
                        } else if (choice == 2) {
                            car.rentCar();
                        } else {
                            car.returnCar();
                        }
                        break;

                    case 4:
                        System.out.println("Available cars: "
                                + countAvailable(0));
                        break;

                    case 5:
                        System.out.print("Enter car ID to search: ");
                        searchCar(sc.nextInt());
                        break;

                    case 6:
                        sortCars();
                        break;

                    case 7:
                        showAvailableCars();
                        break;

                    case 8:
                        saveCars();
                        break;

                    case 9:
                        readCars();
                        break;

                    case 10:
                        System.out.println("Thank you!");
                        break;

                    default:
                        System.out.println("Invalid choice.");
                }

            } catch (InputMismatchException e) {
                System.out.println("Please enter a valid number.");
                sc.nextLine();
            } catch (Exception e) {
                System.out.println("Error: " + e.getMessage());
            }

        } while (choice != 10);

        sc.close();
    }
}
```